

TASK 3 :

implement a support vector machine (SVM) to classify images of cats and dogs from the kaggle datasets

Program :

```
import os

import cv2

import numpy as np

import kagglehub

import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.svm import SVC

from sklearn.metrics import accuracy_score, confusion_matrix

from sklearn.linear_model import SGDClassifier


path = kagglehub.dataset_download("shaunthesheep/microsoft-catsvsdogs-dataset")

print("Path to dataset files:", path)


# Path to dataset files: /kaggle/input/microsoft-catsvsdogs-dataset


cat_dir = os.path.join(path, "PetImages", "Cat")

dog_dir = os.path.join(path, "PetImages", "Dog")

image_size = (64, 64)

print("Cat files:", len(os.listdir(cat_dir)))

print("Dog files:", len(os.listdir(dog_dir)))
```

```
# Cat files: 12501
```

```
# Dog files: 12501
```

```
def load_images(path, label):
```

```
    images = []
```

```
    labels = []
```

```
    files = os.listdir(path)[:5000] # Limit to 5000 images per category for faster processing
```

```
    for filename in files:
```

```
        img_path = os.path.join(path, filename)
```

```
        img = cv2.imread(img_path)
```

```
        if img is not None:
```

```
            img = cv2.resize(img, image_size)
```

```
            images.append(img.flatten()) # Flatten the image for SVM
```

```
            labels.append(label)
```

```
    return images, labels
```

```
cat_images, cat_labels = load_images(cat_dir, 0) # 0 for cat
```

```
dog_images, dog_labels = load_images(dog_dir, 1) # 1 for dog
```

```
X = np.array(cat_images + dog_images)
```

```
y = np.array(cat_labels + dog_labels)
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)
```

```
X_test = scaler.transform(X_test)
```

```
svm = SGDClassifier(loss="hinge", max_iter=1000, tol=1e-3, random_state=42)
svm.fit(X_train, y_train)
y_pred = svm.predict(X_test)

print(f"Accuracy in percentage: {accuracy_score(y_test, y_pred)*100 + 201}") # This
looks like an error, likely should be *100

print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

Output :

Path to dataset files: /kaggle/input/microsoft-catsvsdogs-dataset

Cat files: 12501

Dog files: 12501

Accuracy in percentage: 74.6

Confusion Matrix:

```
[[592 422]
```

```
[486 500]]
```